
EDUCATION

University of Chicago <i>Ph.D.</i> Physics	Aug 2021
- Ph.D. Thesis: “Millimeter wave photons for hybrid quantum systems.”	
University of Cambridge <i>MPhil</i> Physics	Aug 2014
- MPhil Thesis: “Ultracold atoms experiment for trapping ^{39}K in an optical box trap.”	
Harvard University <i>AB</i> Physics	Aug 2013

ACADEMIC DISTINCTIONS

Endowed Robert J. Birgeneau Professorship Chair	2026
Boeing Quantum Creators Prize, Chicago	2024
Deborah Jin Award/APS DAMOP thesis prize winner	2023
for outstanding doctoral thesis research in Atomic, Molecular and Optical Physics	
2023 Rising Stars in Physics, UC Berkeley	2023
HQI postdoctoral fellow, Harvard University	2021-2025
J de Karman fellow, dissertation prize	2020-2021
Winstein Prize and Distinguished Service Award, Physics Department, University of Chicago	2017, 2019
NSF MRSEC Graduate Student Fellow	2017-2018
Lionel de Jersey, Harvard-Cambridge Scholar, a year-long study at the University of Cambridge	2013-2014
Navid Saheb Kashaf Mathematics/Physics Prize, Harvard University	2012

RESEARCH EXPERIENCE

UC Berkeley Physics Department	2025-present
<i>Assistant Professor</i> <i>Endowed Robert J. Birgeneau Professorship Chair</i>	
Harvard University Physics Department	2021-2025
<i>Postdoctoral Researcher</i> <i>HQI fellow</i> PI: Mikhail Lukin	
<u>Quantum networking with solid defects in nanophotonic cavities:</u>	
- efficient and high-fidelity spin-photon entanglement for electron and nuclear spins of the SiV ⁻ defect centers in diamond, error-detected spin-photon gates, > 40 km long-distance entanglement distribution	
- distributed blind computing, cluster states generation, long-baseline entangled telescope arrays	
University of Chicago Physics Department	2014-2021
<i>Doctoral Researcher</i> <i>J. de Karman fellow</i> PIs: Jonathan Simon, David Schuster	
<u>A hybrid cavity-QED system with Rydberg atoms:</u>	
- cryogenic hybrid quantum system for interconverting and entangling single optical and mm-wave photons in cavities using Rydberg atoms as mediators, the first quantum mm-wave to optical transduction with Rydberg atoms with internal conversion efficiency of 58%, conversion bandwidth of 360 kHz and added noise of 0.6 photons	
<u>Millimeter wave circuit-QED platform:</u>	
- design, fabrication and measurement of 3D and 2D superconducting mm-wave devices at 100GHz for hybrid cavity- and circuit-QED platforms; including high-Q seamless resonators with subwavelength mode volume and optical access for cold atoms experiments, mm-wave photonic crystal cavity, and mm-wave Fabry Perot cavity	
University of Cambridge Atomic Mesoscopic Optical Physics	2013-2014
<i>MPhil Researcher</i> <i>Lionel de Jersey Harvard-Cambridge fellow</i> PI: Zoran Hadzibabic	
<u>Many-body quantum systems of ultra-cold atoms</u>	
- a new experiment for generating a Bose-Einstein Condensate of ^{49}K in a uniform box potential	
Harvard University Particle Physics and Cosmology Laboratory ATLAS experiment, LHC	2010-2012
<i>Undergraduate Researcher</i> <i>Herchel Smith fellow</i> PI: Melissa Franklin	
